PassFrame tuition

Speed, Distance & Time

Year 4 Pt 2 — Homework ⏱ 45 min

Worked example — read this through before you start

Worked example: A cyclist travels 45 km in 1.5 hours. What is her average speed?

Method & answer: 45 ÷ 1.5 = 30 km/h.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A car travels 90 km in 1.5 hours. What is its speed?

2. Which formula finds time? A) Distance × Speed B) Distance ÷ Speed C) Speed ÷ Distance D) Speed − Distance

3. A cyclist rides at 16 km/h for 2.5 hours. How far does he travel?

4. A train covers 260 km in 4 hours. What is its speed?

5. Which of these is a sensible speed for a person walking? A) 5 km/h B) 50 km/h C) 150 km/h D) 500 km/h

6. A walker moves at 6 km/h for 1.5 hours. How far does he walk?

7. A car travels 105 km in 1.5 hours. What is its average speed?

8. A bus travels at 36 km/h for 2.5 hours. How far does it travel?

9. Select ALL the speed calculations below that are correct: A) 150 km in 2.5 hours = 60 km/h B) 80 km in 2 hours = 50 km/h C) 126 km in 3 hours = 42 km/h D) 45 km in 1.5 hours = 30 km/h E) 200 km in 4 hours = 40 km/h

10. A car travels 220 km in 4 hours. What is its speed?

11. A runner covers 9 km in 1.5 hours. What is her speed?

12. Which of these speeds is the fastest? A) 48 km/h B) 54 km/h C) 39 km/h D) 51 km/h

13. A coach travels at 46 km/h for 2.5 hours. How far does it travel?

14. A car takes 3.5 hours to travel 245 km. What is its average speed?

15. A cyclist travels 15 km in the first hour and 21 km in the second hour. What is his average speed over the 2 hours?

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A ship sails 315 km in 4.5 hours. What is its speed?

2. A tractor drives at 14 km/h for 2.5 hours. How far does it travel?

6. A bus travels at 38 km/h for 2.5 hours. How far does it travel?

7. A plane flies at 620 km/h for 1.5 hours. How far does it travel? A) 840 km B) 900 km C) 930 km D) 960 km

11. A car travels at 50 mph. Select ALL statements that are TRUE: A) In 2 hours it covers 100 miles B) In 30 minutes it covers 20 miles C) In 3 hours it covers 150 miles D) In 15 minutes it covers 12.5 miles E) In 1 hour 12 minutes it covers 60 miles

12. A train takes 2.5 hours to travel 235 km. What is its average speed?

3. A car travels 310 km in 5 hours. What is its speed? A) 58 km/h B) 60 km/h C) 62 km/h D) 64 km/h

4. A car travels for 45 minutes at 72 km/h. How far does it travel?

5. A train travels 372 km in 4 hours. What is its speed?

8. A cyclist covers 57 km in 3 hours. What is her speed?

9. A car travels for 20 minutes at 90 km/h. How far does it travel?

10. A walker travels at 4 km/h for 105 minutes. How far does he walk?

13. A coach travels at 58 km/h for 2.5 hours. How far does it travel?

14. Which journey has the greatest speed? A) 210 km in 3 hours B) 168 km in 2 hours C) 310 km in 4 hours D) 144 km in 1.5 hours

15. A car travels 105 km in the first 1.5 hours and 175 km in the next 2.5 hours. What is its average speed for the whole journey?

Standard / Advanced — *work through together, answer 5 in the lesson* *15 min (1 min per question)*

1. A car's average speed for a 165 km journey is 55 km/h. How many minutes does the journey take?

2. A train travels 175 km in 2 hours 30 minutes. What is its average speed? A) 60 km/h B) 65 km/h C) 70 km/h D) 75 km/h

3. A plane's speed is 640 km/h. How far does it travel in 45 minutes?

4. A cyclist travels at 32 km/h. How long, in minutes, does she take to travel 8 km?

5. A car travels 260 km at an average speed that is 15 km/h faster than 50 km/h. How long does the journey take? A) 3 hours B) 3.5 hours C) 4 hours D) 4.5 hours

6. A lorry travels the first 120 km of a journey at 60 km/h, then the next 175 km at 70 km/h. How long does the whole journey take?

7. Two trains leave stations 400 km apart at the same time, travelling towards each other. One travels at 70 km/h, the other at 90 km/h. After how many hours do they meet?

8. A car travels 400 km at an average speed of 80 km/h. Select ALL of the following that are TRUE: A) The journey takes 5 hours B) Increasing the speed to 100 km/h would make the journey take 4 hours C) In 2 hours the car has covered exactly half the journey D) At 80 km/h, the car covers 20 km every 15 minutes E) The journey would take 8 hours at 50 km/h

9. A cyclist rides 48 km at 24 km/h, then rides back the same route at 16 km/h. How long does the whole round trip take?

10. A car leaves at 07:40 and travels 225 km at an average speed of 90 km/h, including a 20-minute stop. At what time does it arrive? A) 10:00 B) 10:20 C) 10:30 D) 10:40

11. A ship sails 70 km at a speed of 35 km/h, then sails a further 120 km which takes 3 hours. What is the ship's average speed for the whole journey?

12. A runner completes a 24 km course: the first 18 km at 12 km/h, and the remaining 6 km at 8 km/h. How long does she take in total?

13. A car and a motorbike both leave the same point at the same time, travelling in the same direction. The car travels at 75 km/h and the motorbike at 95 km/h. How far apart are they after 2 hours? A) 30 km B) 35 km C) 40 km D) 45 km

14. A cargo ship travels 450 km. For the first 280 km it averages 70 km/h. For the whole journey to take 8 hours, at what average speed must it complete the final 170 km?

15. A train's journey has three legs: 80 km at 80 km/h, 105 km at 70 km/h, and 60 km at 90 km/h. What is the train's average speed for the whole journey, to the nearest whole km/h?

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